Attitudes Towards Generative Artificial Intelligence in the Ultimatum Game

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Abstract:

This paper aims to investigate the extent to which human attitudes towards artificial intelligence (AI) impact personal and societal welfare through economic game experiments. Using the classic ultimatum game, we can test how humans alter their decision-making processes when they are aware they are matched with an AI player. Existing research suggests that humans differ in their attitudes towards AI—some believe it accurately imitates human behavior and can be undetectable, others fear and refuse to cooperate with it. The literature suggests that we can expect to see stronger preferences for fairness in AI players and weaker preferences for fairness in human proposers against AI responders, though this is partially mitigated against AI responders with human partners receiving the earnings of the game. Overall, we find results similar to those of the literature, with the added finding that positive attitudes towards AI are correlated with less fairness when bargaining with it. This study aims to contribute to a greater assessment of the always-changing landscape of human-AI relationship dynamics, while calling attention to the need for more research in this area to better understand the wider scope of its societal implications.

Introduction:

In the world of ever-improving technology, artificial intelligence (AI) is advancing at breakneck speeds. With large language models (LLMs) constantly developing, training, and evolving, the scope of realistic applications has grown to include many aspects of human daily lives—including those regarding efficient decision-making processes. This proliferation of AI use has caused an influx of human-AI interaction, which will only continue to rise with further advancements in the AI field. This raises questions on how these interactions differ from how people treat other humans. This paper seeks to analyze human behavior when bargaining with AI; specifically, we aim to observe changes in preferences for fairness and their effects on efficiency and welfare. These changes will occur based on whether people are bargaining with an AI, are aware they are bargaining with AI, whether the AI is attached to a human partner, and based on the individual's attitudes towards AI. We utilize OpenAI's 4th generation Chat Generative Pre-trained Transformers (GPT-4) large language model, analyzing its impact on the ultimatum game. This is a well-studied game theory experiment, which pairs participants, asking one—the proposer—to divide a sum of money between the two, and the other—the responder—to either accept or reject this allocation. Proposers could either attempt to maximize their own payoff, as theory predicts, or balance the fairness of their division. Responder payoffs are at the mercy of the proposed splits and contingent on their decision to accept or reject. Analyzing these proposal patterns at the level of the individual may reveal their own underlying motivations. Proposers fall into categories of either overtly selfish, equitably fair, or overly altruistic. Responders however, are more difficult to understand, as their decisions are dependent upon both their preexisting attitudes and the proposal. This experiment is repeated to study the

development of trust and reciprocity in the proposer-responder relationship across multiple rounds.

The introduction of programmable AI agents into the ultimatum game adds new elements to these decisions. Rather than random partnership with a human player, participants are matched with an AI agent in either of the roles. By inserting the AI with no other changes to procedure, we directly observe how human behavior changes given the presence of AI. This change will be studied across several treatments, with human participants and AI agents taking on the role of both proposers and responders. Other treatments include differences in information— knowledge of AI participants being introduced, and toggling of associated payoffs going to human recipients. Understanding trends in fairness preferences associated with human-AI interaction is useful on several fronts: first and foremost, we seek to understand how humans react when interacting with AI instead of humans in order to refine AI models to make these interactions run smoother. Moreover, discrepancies between human and AI behavior in the data can help prevent instances where AI may not act in human's best interests, and help mitigate instances where AI actions were taken incorrectly. Identifying the reasoning behind potential differences in theoretical and empirical behavior in these interactions can also help us learn more about society and individuals, e.g. their preferences, underlying motivations, and logical reasoning. These preferences were revealed in a post-experiment survey completed by human participants. Furthermore, this can help train AI models to more accurately represent human behavior in the future, aiding in other research areas.

Literature Review:

Previous research endeavors regarding the integration of AI into economic games spurred our experimental pursuits. Specifically, our investigation was motivated by existing work concerning the use of generative AI in public goods game experiments (Villa 2024). This work used basic methodologies i.e. difference of means tests and simple linear regression to analyze the impact that knowledge of AI agents in the public goods game on the welfare maximizing decisions made by the human participants. Villa concludes that in the case of the public goods game, knowledge that one member of the group was AI had no significant impact on neither the decisions made nor the overall payoff of human participants. However, he does find that AI agents acted more altruistically than humans. Capraro, et al., (2024) corroborates this finding. They tested the various LLMs' abilities to capture and reproduce qualitative behavioral patterns, falling under the classifications of selfish, fair, and altruistic. They find that GPT-4 was the only model that could perform correctly, but even then, it "consistently underestimates self-interest and inequity-aversion, while overestimating altruistic behavior" (Capraro, et al., 2024). We will discuss whether we find similar altruism in our results. Regarding the insignificance of the presence of AI in terms of human behavior in the public goods game, the effect of the AI may have been crowded out by human behavior given that there was only one AI agent in each group of four. The ultimatum game is a much more direct interaction. This is not to say that the public goods game was not a useful experiment; it merely observed a different type of situation, where an AI agent was placed among many human actors. Our experiment simulates a situation where a human and AI agent are interacting one on one.

Other recent studies focus mainly on the human side of these interactions. In their seminal paper, Dvorak, et al., (2024) evaluated trends in welfare through several other relevant economic games: the prisoner's dilemma, coordination game, binary trust game, stag hunt game,

and finally, the ultimatum game. In certain treatments in these experiments participants were given the option to delegate decisions to an AI partner. The authors found that participants consistently delegated to the AI when given the opportunity, which only makes it more clear that human-AI interactions will be extremely important in the future; people want AI to perform tasks and make decisions for them, so we need to understand these interactions and how they affect people's welfare. The study concluded that in social interactions, generative AI causes humans to make welfare-reducing decisions . Moreover, they found that participants were unable to identify when they were playing against AI in opaque delegation treatments, and that personalization of the AI to match a particular human's preferences did not restore welfare losses. In combination, these results are worrying for the future of human-AI interaction, as a lack of transparency when using AI could cause its use to go unnoticed while inducing welfare-reducing behavior in humans. People do not seem to like interacting with AI either. As put forth by Erlei, et al., (2022), humans may willingly choose lower welfare in order to avoid bargaining with AI systems. Using an extended ultimatum game with voluntary opt-outs at any stage, as well as knowledge of an AI agent being the other participant, they found that human subjects acting as the responder will choose a lower payoff in order to stop interacting with the AI agent (Erlei, et al., 2022). This is another troubling result. Humans will act in a way that reduces welfare when interacting with AI, but will also sacrifice welfare in order to avoid interacting with them in the first place. It is thus essential that we understand the motivations behind this welfare reducing behavior so that a solution can be found.

Ozkes, et al., (2024) tried to isolate some of these effects in their experiments by having different AI algorithms play the ultimatum game with and without a human beneficiary. They found similar preferences to Erlei, et al., (2022) for interacting with humans rather than AI, but

their results differ from Dvorak and Erlei, in that they found no evidence that responders behaved differently when playing against AI. They do find evidence that when acting as the proposer, humans will give higher demands to AI when there is no human beneficiary. However, they find that given a human beneficiary, human proposers will forgo higher payoffs. Similar results were found by von Schenk, et al., (2023) who conducted a meta-analysis of experiments utilizing a mixed human and AI subject pool, focusing on the disparities in experimental instructions for both types of participants. It was found that across economic games with knowledge of AI partners, humans maintained the greatest fairness preferences against fellow humans, dropping significantly when playing against AI with a human beneficiary, and dropping even further against AI with no human beneficiary. This is the result we anticipate in our experiments.

Benndorf, et al., (2021), also analyzed social preferences of individuals in the ultimatum game where they are aware their counterparts are AI. Their goal was to eliminate social preferences and observe outcomes to see if they converge to theory. They designed an experiment in which humans and computers alternate roles in subsequent rounds of the ultimatum game, with randomly assigned matchups, and the computer perfectly inheriting the behavior of the human from the previous round. These behaviors were classified as selfish, fair, or altruistic, depending on the proposer's division of 10 monetary units, giving the responder 0-3, 4-7, and 8-10, respectively. They find that responders became more lenient towards more selfish demands, but that proposer did not become more aggressive against computer responders, thus their behavior does not converge to a trembling-hand perfect equilibrium i.e. deviation from theory cannot be attributed to slips in decision-making under game theoretical selfish preferences. They suggest that proposers found difficulty in figuring out computer behavior, and

that learning interacts with social preferences in the game to make convergence to theory very slow. This contrasts with the previously mentioned experiments that found much more aggressive proposer behavior against AI, especially with no human beneficiary. The alternating waves of computer proposers and responders in this experiment, while an interesting design, is not what we employ. Therefore, we expect results similar to other experiments of similar design to us. However, the Benndorf experiment provides more evidence that the ultimatum game will not converge to theory, even with attempts to eliminate social preferences.

One aspect of interaction with AI that we did not manipulate was behavioral input i.e. changing the way the AI acts, which would be most easily altered using prompt engineering. Some other studies have looked into this. Guo (2023) focused on the prisoner's dilemma and ultimatum game, prompting GPT-4 with the game rules as well as logical reasoning behind each of the choices. These reasonings included motivations for the three previously distinguished categories of behavior. Guo ran both sets of experiments with combinations of behaviorally programmed GPT-4 against one another, and asked the AI to reveal its underlying logic along with its decision. He found that GPT-4 weakly replicated behavioral patterns suggested by the input preferences, and provided data confirming the existence of patterns supporting the effectiveness of behavioral inputs. Kitadai, et al., (2023), investigated the impact of different experimental settings and game variations on AI decision-making. The authors used GPT-3.5 playing against itself, and found that three variables have the most impact on the effectiveness of the AI's decisions: LLM instruction, reiteration of real monetary payoffs, and output randomness, commonly referred to as "temperature." When these variables were left unspecified, the resulting payoffs centered around the mean of the dataset, which was extremely close to that of human behavior. However, if the AI was instructed regarding these variables—for instance,

told to maximize its own payoff and be as greedy as possible—it had significant effects that swayed the resulting data on either side of the mean, and thus, away from typical human behavior (Kitadai, et al., 2023). We did not specify these variables in our experiment, instead leaving them to default settings. This could be an interesting research topic in the future, as the use of AI in various applications will certainly be tailored to their specific purposes.

The level of information given to an AI agent is also an important factor. Castro, et al., (2023) acknowledged the benefits of AI and GPT-4 in general, but found future societal issues could stem from the AI's lack of perfect preference matching: that is, not acting in a way that exactly matches the user's individual motivations and preferred payoffs. They conclude that with a lack of information, AI will fail to perfectly match individuals' societal preferences, which results in significant decision differences in later rounds, impacting overall welfare of the user and society. In order to mitigate this, they suggest giving as much detailed information to the AI as possible, in order to match social preferences (Castro, et al., 2023). In our experiment, we prompt the AI with the same instructions each round in an effort to reduce the noise associated with insufficient information in prompts.

Experimental Design:

We employ a standard ultimatum game; every round participants divide ten dollars into whole-dollar splits across ten rounds of play with repeated matching. The human subject pool consisted of 299 participants distributed into five treatments, with around 50 proposers and 50 responders per treatment. Treatment one was a baseline human to human group, with no AI agents. The second treatment was human proposer to AI responder with 'no info', i.e. the human participants were not told that they were playing against AI agents. Treatment three was AI

proposer to human responder with no info. Our fourth treatment was human-AI with info but no human partner—that is, no human was receiving the earnings made by the AI throughout the game. Finally, our fifth treatment was human-AI with info and human partners, where a human does not make decisions but receives the earnings of the AI agent they are paired with. The participants were randomly chosen to be either proposers or AI-responder partners in this treatment.

Table 1: Treatment groups				
Treatment 1	Treatment 2	Treatment 3	Treatment 4	Treatment 5
Human proposer Human responder	Human proposer AI responder (no info)	AI proposer (no info) Human responder	Human proposer AI responder (informed, no partner)	Human proposer, AI responder (informed, human partner)

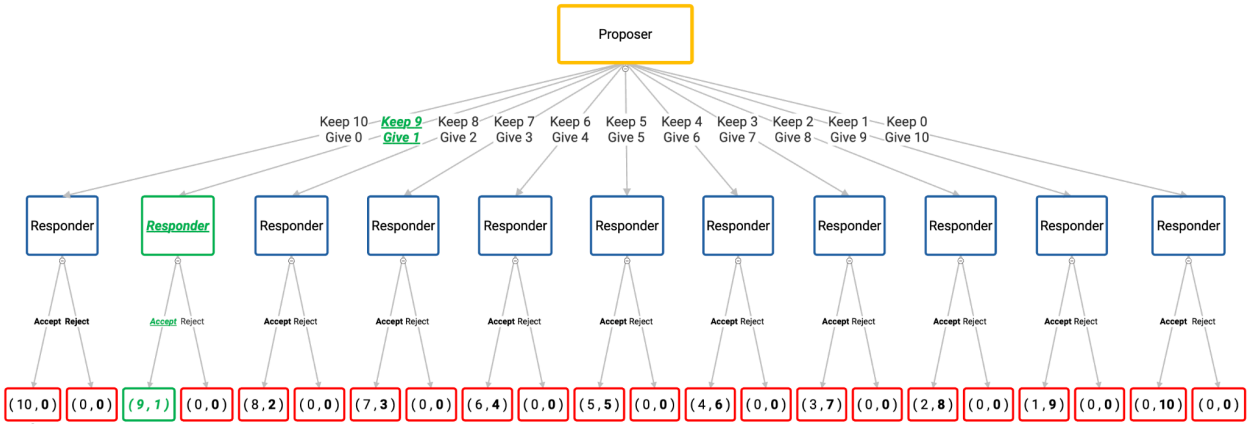
All subjects were undergraduate or graduate students at William and Mary. The data were collected across 25 sessions throughout the fall of 2024 and the spring of 2025, and were conducted by Professor Lisa Anderson, Professor Rob Hicks, and students in Professor Anderson's class. During the sessions, subjects were seated with dividers up to conceal the choices of other participants around them, where they assigned a random ID number to be used for the game to make their choices anonymous. Professor Anderson would then read out the instructions to the game, which are available in our appendix, and the participants would go through the ten rounds of the game on the Veconlab website of the University of Virginia (http://veconlab.econ.virginia.edu), using their pre-designed ultimatum game.

For treatments three through five, a separate room with students from Professor Anderson's class ran a python script on one computer for each AI agent in the experiment that prompted an instance of GPT-4 with the instructions to the game and automatically processed

input and output data between the human participant and the AI, logging the results at the end.[1] Each GPT-4 agent was prompted with the same game instructions as humans, pertaining to its role in the experiment, with specific identifiers allowing each round of each experiment to be recorded. In the fifth treatment, the human partners to the AI responders were placed in the room with the computers running the scripts, so they could see what their AI partner was doing.

After they concluded the ten rounds, the participants were given a survey to collect demographic data and information about their attitudes towards their own trustworthiness and their attitudes towards AI. Participants were then given 50% of their cumulative earnings from the game, on top of a flat rate of five dollars each that was distributed prior to beginning the game. Real monetary payoffs helped to ensure participants made meaningful decisions and the post-game survey gave us more qualitative and quantitative data to compare associated subjects' decisions across.

Theoretical Model and Predictions:



[1] This was a change from our setup in treatment two during the fall 2024 semester, where a student from Professor Anderson's class would have to manually input each decision into the Python interface with the AI, and manually submit the output to Veconlab.

The ultimatum game is solved for equilibrium using the subgame perfect Nash equilibrium (SPNE). By implementing backwards induction starting from the last round of the game, the responder should accept any proposer demand that leaves a positive amount for the responder, as it would be better for them than rejecting and receiving zero dollars; the proposer, understanding this, should demand nine dollars and offer one dollar, giving themselves the maximum division that will be accepted by the responder. Applying this logic to every round in reverse order, the cumulative earnings of the responder should be ten dollars, and the proposer should end with ninety. This is also an efficient outcome, as all available units of currency are allocated to either the proposer or responder. Experimental data reveals that people do not act according to theory. In general, offers converge to somewhere between 50-50 and a 60-40 division in favor of the proposer (Güth 2013).

In our experiment, we test for treatment differences explained by disparate fairness considerations between AI and humans, as well as tendencies for humans to alter their aggression given the knowledge of AI opponents and human partners to said opponents. Our predictions align with what the nascent literature in this area has shown: humans will be more aggressive against AI opponents with no human partner, while preserving preferences for fairness when a human is receiving the earnings from the game. This means we should see the highest proposer demands in treatment four with informed human proposers against AI responders with no human beneficiary, and higher demands in treatment five than treatments one and two, but lower than treatment four, as the human beneficiary elicits altruism from the human proposers. On the AI side, we predict a stronger preference for fairness, based on previous experiments. For example, in the public goods experiment conducted by Villa (2024), GPT-4 was more altruistic than humans. As a corollary, we expect more rejections in the AI responder

treatments, due to their fairness preferences and the higher demands we expect when the human proposers are informed, resulting in less efficient outcomes. In the AI proposer treatment 3, the preference for fairness should lead to very even divisions and high levels of efficiency. We will be using simple linear regressions, usually on dummy variables to measure treatment differences pairwise on all five treatments. For our investigation on the effect of attitudes towards AI, we will also use dummy variables, but within treatments instead of between. These will be based on answers to our survey questions. The exact formulation will be discussed in the next section. We predict that positive attitudes will foster trust and correlate with fairer proposer demands and more efficient outcomes i.e. higher cumulative earnings.

Results:

We begin with an overview of cumulative earnings, proposer demands, and acceptance rates. It is immediately apparent that the informed treatments with AI responders produced less fair results, as proposer demands were higher, acceptance rates were lower, and the cumulative earnings gap was larger. Treatment three with AI proposers was the most fair with an average demand of almost exactly five dollars and the highest average cumulative earnings. This made it the most efficient outcome. Delving deeper into these treatment differences reveals more on the motivations behind the behavioral differences of the humans across treatments.

Using simple linear regression on dummy variables to analyze each pair of treatments, we found that all differences between treatments for proposer demands were significant (p = 0.000) except for the difference between treatments one and two, which was both practically and statistically insignificant, with a coefficient of -4 cents and a p-value of 0.437. This regression does not follow previous literature showing preferences for fairness in AI agents, however, looking at the acceptance rates across the range of proposer demands indicates that the AI agents were indeed disinclined towards unfair demands relative to human responders (see

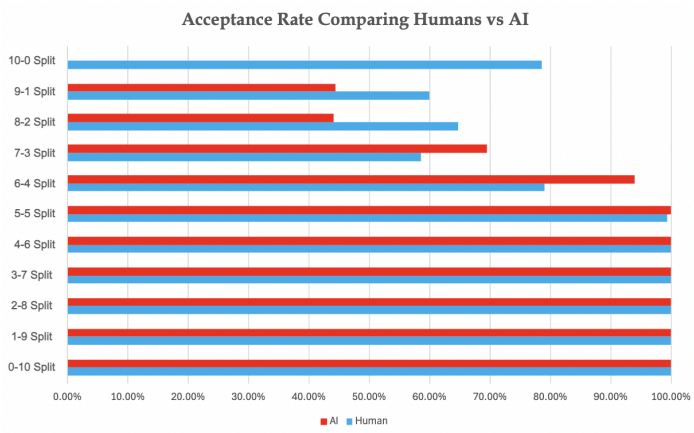


Table 2: Descriptive statistics (averages)					
	Treatment 1	Treatment 2	Treatment 3	Treatment 4	Treatment 5
Proposer cumulative earnings ($)	47.23	47.58	48.83	49.47	49.33
Responder cumulative earnings ($)	44.68	45.69	48.55	37.64	42.33
Proposer demand ($)	5.29	5.25	5.01	5.94	5.60
Acceptance rate (%)	91.91	93.27	97.14	87.12	91.67

figure). Average proposer demands between these treatments were still similar due to the higher demands being a very small percentage of each treatment. Looking at the other treatments, we can see that the difference between human proposers in treatment one and AI proposers in treatment three is statistically significant, with a near 30 cent difference; this result corroborates

our predictions and the fairness preferences that AI agents have exhibited in past experiments. For the AI treatments with information, there was a 68 cent difference with no human partner and a 34 cent difference without, when compared to the uninformed AI responder treatment. Similar results follow from the human to human treatment without information because of the similarity between treatments one and two. This helps to clarify the confounding reasons behind human proposer behavior against AI. The difference is motivated partly by the knowledge that they are playing against AI, hence the 68 cent difference in treatment four; the human proposers think they can extract more payoff out of the AI agent. This difference is tempered by the altruism introduced in treatment five with a human beneficiary. However, this altruism does not completely negate the self-serving motivation that made the human demand more in treatment four, indicated by the remaining 34 cent difference compared to treatment five. Overall, our results fit very closely with the existing literature that motivated our predictions.

Table 3: Treatment differences in variables

	T1 vs. T2	T1 vs. T3	T1 vs. T4	T1 vs. T5	T2 vs. T4	T2 vs. T5	T4 vs. T5
Proposer demand ($)	Coeff = -0.04 P = 0.437	Coeff = -0.28 P = 0.000	Coeff = 0.64 P = 0.000	Coeff = 0.30 P = 0.000	Coeff = 0.68 P = 0.000	Coeff = 0.344 P = 0.000	Coeff = -0.34 P = 0.000
Unfair demands (% likelihood)	Coeff = -6.19 p = 0.000	Coeff = -13.63 p = 0.000	Coeff = 14.64 p = 0.000	Coeff = 9.50 p = 0.000	Coeff = 20.84 p = 0.000	Coeff = 15.70 p = 0.000	Coeff = -5.10 p = 0.010
Acceptance rate (%) on unfair demands	Coeff = -31.28 P = 0.000	Coeff =-64.61 P = 0.06 *only 2 obs.	Coeff = -5.69 P = 0.261	Coeff = 1.71 P = 0.751	Coeff = 25.60 P = 0.000	Coeff = 32.99 P = 0.000	Coeff = 7.40 P = 0.091
Proposer cumulative earnings ($)	Coeff = 0.34 p = 0.733	Coeff = 1.60 p = 0.155	Coeff = 2.24 p = 0.197	Coeff = 2.09 p = 0.199	Coeff = 1.89 p = 0.244	Coeff = 1.75 p = 0.256	Coeff = -0.14 p = 0.947
Responder cumulative earnings ($)	Coeff = 1.01 p = 0.443	Coeff = 3.87 p = 0.003	Coeff = -7.04 p = 0.000	Coeff = -2.35 p = 0.227	Coeff = -8.05 p = 0.000	Coeff = -3.36 p = 0.042	Coeff = 4.69 p = 0.021

Looking at the treatment differences for cumulative earnings and unfair demands (defined as demands greater than $7) gives us more insight on how the unfairness between treatments affects proposers and responders differently. We again utilize dummy variables to observe treatment differences. As the proposer demands suggest, the dichotomy of fairness relative to the baseline human to human treatment was split between the more fair uninformed AI treatments, two and three, and the informed AI treatments, four and five, with four being the least fair because of the lack of a human beneficiary. This aligns with our expectations that without knowledge of the AI the humans would behave similarly to the baseline and that the AI's stronger fairness preferences would push the outcomes to a more fair result, hence the lower percentage of unfair demands and lower acceptance rate on these unfair demands. We also see evidence for our prediction that informed treatments would see more unfair demands as human proposers attempted to exploit the AI for their own profit. The acceptance rate on unfair offers reveals that the AI responders in treatments four and five accepted unfair offers at a substantially higher rate than in the uninformed treatment, and not at a statistically significant different rate compared to the baseline human treatment. This suggests human proposers who were aware that they had AI opponents were emboldened to keep demanding unfair amounts, causing the AI responders to yield to these unfair demands over the course of the game. This theory also aligns with the higher percentage of unfair demands. We also see the anticipated lower levels of unfair demands in treatment three due to the AI's altruistic tendencies.

Furthermore, cumulative earnings for the proposers and responders tells us that proposers did not have a statistically significant difference across treatments, but responder cumulative earnings varied greatly. This indicates that the burden of the unfairness falls squarely on the

responders, as the lower acceptance rates are offset by the higher demands on the proposer side to keep earnings relatively equal. Relative to the baseline, the responder earnings were nearly

Table 5: Average rejected demand and change in demand the round after rejection					
	T1	T2	T3	T4	T5
Rejected demand ($)	7.05	7.32	5.35	7.68	7.94
Change in demand ($)	-1.57	-1.83	-0.93	-1.62	-2.09

four dollars higher in treatment three, seven dollars less in treatment four, and two dollars less in treatment five. This has implications for anyone using a default settings large language model with fairness preferences in bargaining type situations. A human will be able to encroach on the benefits that the AI agent is attempting to secure if they know they are interacting with one. This is in spite of the fact that proposer cumulative earnings increased only marginally with this more aggressive playstyle, while responder cumulative earnings decreased substantially, reducing efficiency greatly in addition to being more unfair.

Continuing our investigation of fairness, we examined the proposer reaction to a rejection, comparing across treatments the change in demand in the round immediately following a rejection. Treatment three's tight distribution of proposals meant the only rejections were either from slight deviations in the proposal or random decisions by the human responders to reject an even split. The more interesting results lie in the treatments with human proposers. Demands in treatments one and two drop to around $5.50 following a rejection, while in the informed AI responder treatments demands drop to around six dollars. The different average rejected demands for treatments between these two groups were very significant with p-values all under 0.02. We suspect this behavioral difference in the AI is mostly due to the more extreme quantity and

magnitude of unfair offers in these treatments, but further controlled testing should be done to determine this, perhaps in one-shot games to eliminate the desensitizing effects of repeated trials.

Treatment differences in the change in demand following a rejection were largely statistically insignificant. The only statistically significant difference was in treatment four vs treatment five, where proposer demands went down by 47 cents more in treatment five. This result conveys that human proposers did not want to retreat from their high demands when they thought they were playing against human responders or AI responders with no human partners, compared to when they were playing against AI responders with human partners. This furthers the idea that a combination of selfish tactical play, and altruism motivated the human proposers. We posit that for treatments one and two, they did not want to back off of their offers as much because they believed they could get an acceptance out of a demand drop that was less steep, indicating a selfish tactical motivation. In treatment four, they did not want to reduce their offer as much because they did not care about the opponents payoff given that it was only an AI agent, signalling selfishness. In treatment five, the drop was much steeper than treatment four because they knew a human was receiving the earnings, signalling altruism. However, this explanation of altruism is inconsistent with the smaller demand drops in treatments one and two. This could potentially be explained by humans reacting more severely to a rejection from AI with a human beneficiary, perhaps expecting that they need to lower their demand more than against a human.

Table 4: Treatment differences: change in demand one round after rejection

	T1 vs. T2	T1 vs. T4	T1 vs. T5	T2 vs. T4	T2 vs. T5	T4 vs. T5
Rejected demand ($)	Coeff = 0.27 p = 0.174	Coeff = 0.63 p = 0.000	Coeff = 0.89 p = 0.000	Coeff = 0.36 p = 0.019	Coeff = 0.62 p = 0.001	Coeff = 0.26 p = 0.099
Change in demand ($)	Coeff = -0.26 p = 0.343	Coeff = -0.05 p = 0.823	Coeff = -0.52 p = 0.095	Coeff = 0.21 p = 0.208	Coeff = 0.26 p = 0.330	Coeff = -0.47 p = 0.019

The differences between the uninformed and informed treatments is harder to explain, as the expectation would be that proposers would maintain unfair demand pressure on responders they knew were AI, but the data do not reflect that. However, the drops in demand returning to values of around $5.50 and $6.00 for uninformed vs. informed treatments point to the previous hypothesis of selfish demands in the informed treatments, despite the larger drops in demand in the informed. In other words, the demands are dropping more in the informed treatments but they start so much higher that the overall result is still sustained pressure of unfair demands.

The survey data collected at the end of the experiment could give us valuable information about how people's stated preferences towards AI affect their decisions in interactions with it. Survey answers were made on a scale of one to five, with one being strongly agree and five being strongly disagree. For each question, two dummy variables were constructed by dividing the answers into positive and negative attitudes. Each dummy variable equals one if the answer is greater than or less than three, depending on the phrasing of the question. For example, for the statement 'AI is a force for good in society', the positive dummy variable equals one if the answer is less than three (agree), and the negative dummy variable equals one if the answer is greater than three (disagree). This was done to isolate positive and negative effects and exclude neutral answers. This allows us to see if positive or negative attitudes have a greater effect than the other on human proposer behavior. We looked specifically at a regression of the variables on proposer demand. In order for an effect from one of these questions to have a useful interpretation, there must be a statistically significant coefficient within treatments four and/or five, and no underlying significant coefficient in treatments one and/or two. If these criteria are met, then we can conclude that a human proposer's attitude towards AI affected their play when

informed they were playing against an AI agent. Many of the questions were irrelevant or had no statistical significance, but four of them had useful results.

Table 6: Effect of positive attitudes on proposer demands within treatments				
	Treatment one	Treatment two	Treatment four	Treatment five
AI is a force for good (agree)	Coeff = 0.30 p = 0.088	Coeff = 0.13 P = 0.217	Coeff = 0.53 P = 0.001	Coeff = 0.23 P = 0.277
Interested in using AI in daily life (agree)	Coeff = 0.24 P = 0.099	Coeff = -0.02 P = 0.780	Coeff = 0.39 P = 0.001	Coeff = 0.77 P = 0.000
AI has positive effects on wellbeing (agree)	Coeff = 0.03 P = 0.863	Coeff = 0.03 P = 0.587	Coeff = 0.69 P = 0.000	Coeff = 0.16 P = 0.251
AI systems can make many errors (disagree)	Coeff = 0.28 P = 0.238	Coeff = 0.02 P = 0.919	Coeff = -0.75 P = 0.071	Coeff = 0.95 P = 0.003

Table 7: Effect of negative attitudes on proposer demands within treatments				
	Treatment one	Treatment two	Treatment four	Treatment five
AI is a force for good (disagree)	Coeff = -0.16 P = 0.278	Coeff = -0.05 P = 0.551	Coeff = -0.01 P = 0.923	Coeff = -0.61 P = 0.000
Interested in using AI in daily life (disagree)	Coeff = -0.07 P = 0.627	Coeff = 0.15 P = 0.041	Coeff = 0.12 P = 0.273	Coeff = -0.78 P = 0.000
AI has positive effects on wellbeing (disagree)	Coeff = 0.29 P = 0.173	Coeff = -0.17 P = 0.075	Coeff = -0.15 P = 0.437	Coeff = 0.20 P = 0.313
AI systems can make many errors (agree)	Coeff = -0.05 P = .708	Coeff = -0.22 P = 0.013	Coeff = 0.50 P = 0.000	Coeff = -0.14 P = 0.368

Looking at p-values within the 0.05 boundary for statistical significance, we find the results we were looking for in all of these questions. Some of them have statistically significant or near significant coefficients in treatment two, but these are all in the opposite direction of the coefficients in treatments four and five, which only strengthens their case for having an effect on proposer behavior. Many of these variables were significant only in treatment four or treatment five but not the other, which calls into question their usefulness. However, after finding the specific survey questions with useful interpretations we constructed an index for attitude towards

AI; this index was a linear combination of the answers, from one to five, to these four questions, divided by four, adding or subtracting based on whether the question had a positive or negative phrasing. The index aims to use each question to cover any deficiencies that each individual question may have. The coefficients on the index match the individual effects of all of the questions, which all indicate that the more positive the attitude towards AI the higher the proposer demands will be in treatments with informed human proposers against AI responders. This manifests as negative coefficients in the index because the higher numbers indicate disagreement.

Table 8: Effect of index on proposer demands				
	Treatment one	Treatment two	Treatment four	Treatment five
Index effect	Coeff = -0.09 P = 0.392	Coeff = -0.004 P = 0.948	Coeff = -0.25 P = 0.003	Coeff = -0.31 P = 0.000

The regressions imply that people that are more optimistic towards AI will try to capitalize on its involvement in the game to maximize their payoff. To account for nonlinearity in the regression on the index, we included a quadratic term, but found this to be insignificant, meaning linearity is reasonably accurate in the coefficients. This was a surprising result, as we predicted that positive attitudes would correlate with lower proposer demands due to trust, but we find that people instead take AI as an opportunity for their own gain.

Conclusions:

This paper analyzed how human decision-making is influenced by knowledge of partnership with AI players within an ultimatum game experiment. Specifically, we looked at measures of fairness and attitudes towards AI. This research adds to the rapidly growing expanse of literature on human-AI interaction in the context of social sciences, seeking to study how

decision-making changes with this knowledge and to what extent it does change. Our results confirm what the wider literature has evinced; humans are most concerned with fairness when playing other humans, are less generous against AI with human beneficiaries, and are the least generous against AI with no human beneficiary. Our investigation of the effect of attitudes towards AI revealed more aggressive demands associated with positive attitudes, which warrants more investigation as human-AI interaction increases in the future. Understanding how people's attitudes affect behavior is important as both the sophistication of AI and people's opinions towards it shift in the future. Although this research represents significant efforts towards a better understanding of the relationship between humans and AI, it is not without its limitations. Specified funding limited our experimental design—particularly, we used a financially accessible LLM, and we were limited in our capacity to incentivize and reward participants. This paper itself was limited by general length guidelines and time-sensitive deadlines. While it already takes place among a wider body of research, ultimatum game research could in the future take place over a much longer period of time, aimed at tracking the improvements in LLM abilities. Future research could also delve into AI interactions in other economic games, as well as different instances of measuring welfare, such as socially accurate preference matching. Prompt engineering and behavioral input are other important areas for future studies, as most interactions between humans and AI in bargaining situations will utilize AI that is tuned to that specific purpose.

References:

Benndorf, V., Große Brinkhaus, T., & von Siemens, F. A. (2021). Ultimatum Game Behavior in a Social-Preferences Vacuum Chamber. *CESifo Working Papers*, *9280*. https://www.cesifo.org/DocDL/cesifo1_wp9280.pdf

Capraro, V., Pizziol, V., & Di Paolo, R. (2024). Assessing Large Language Models' ability to predict how humans balance self-interest and the interest of others. *Social Science Research Network*. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4517639

Castro, F., Gao, J., & Martin, S. (2023). *Human-AI Interactions and Societal Pitfalls*. Cornell University. https://arxiv.org/abs/2309.10448

Dvorak, F., Stumpf, R., Fehrler, S., & Fischbacher, U. (2024). Generative AI Triggers Welfare-Reducing Decisions in Humans. *ArXiv (Cornell University)*. https://doi.org/10.48550/arxiv.2401.12773

Erlei, A., Das, R., Meub, L., Arnand, A., & Gadiraju, U. (2022). For What It's Worth: Humans Overwrite Their Economic Self-interest to Avoid Bargaining With AI Systems. *CHI '22 Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems*, *113*, 1–18. https://dl.acm.org/doi/pdf/10.1145/3491102.3517734

Guo, F. (2023). *GPT in Game Theory Experiments*. Cornell University. https://arxiv.org/abs/2305.05516

Güth, W., & Kocher, M. G. (2013). More than Thirty Years of Ultimatum Bargaining Experiments: Motives, Variations, and a Survey of the Recent Literature. *SSRN Electronic Journal*, *35*. https://doi.org/10.2139/ssrn.2319871

Kitadai, A., Tsurusaki, Y., Fukasawa, Y., & Nishino, N. (2023). Toward a Novel Methodology in Economic Experiments: Simulation of the Ultimatum Game with Large Language Models. *2023 IEEE International Conference on Big Data*, 3168–3175. https://ieeexplore.ieee.org/document/10386678

Ozkes, A. I., Hanaki, N., Vanderelst, D., & Willems, J. (2024). Ultimatum Bargaining: Algorithms vs. Humans. *Economics Letters*, *244*. https://www.sciencedirect.com/science/article/pii/S0165176524004634?via%3Dihub

Villa, P. (2024). How Human Decisions are Affected by Knowledge of Artificial Intelligence.

von Schenk, A., Klockmann, V., & Kobis, N. (2023). Social Preferences Toward Humans and Machines: A Systematic Experiment on the Role of Machine Payoffs. *Association for Psychological Science*, *20*(1), 165–181. https://doi.org/10.1177/17456916231194949

Appendices:

Instructions for All Treatments:

Rounds and Matchings: The experiment consists of 10 rounds. Note: You will be matched with the same person in all rounds.

Decisions: One of you will be designated as a Proposer (initial decision-maker) who will begin the round by proposing a division of an amount of money, $10.00. The other person, designated as responder, is told of the proposed division and must either accept it or not.

Proposer-specific Instructions:

Role: You have been randomly assigned to be a Proposer (or initial decision-maker) in this process, and you will begin by suggesting amounts of money for each of you that sum to $10.00.

Responder-specific Instructions:

OR Role: You have been randomly assigned to be a Responder in this process. The other person (proposer) will begin by suggesting amounts of money for each of you that sum to $10.00.

Earnings: If the responder accepts the proposed division, then each person earns their part of the proposed division. If the responder rejects, both earn $0.00.

Instructions Summary: Please remember that you will be matched with the same person in all rounds. The proposer (initial decision-maker) in each pair suggests a division of the $10.00; this proposal determines earnings for the round if it is accepted. A rejection produces earnings of $0.00 for each person. If the proposal is rejected, earnings are $0.00 for both proposer and responder. There will be 10 rounds, and you are always matched with the same person.

Special Earnings Announcement:

Your cash earnings in this experiment will be 50% of your cumulative earnings at the end of the experiment.